

Agenda

- Last time
 - Independence, Pair-wise independence
- Today
 - Covariance
 - Inequalities

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Mutual independence

This definition for n variables is hard to comprehend, see example next

- n continuous random variables are mutually independent iff
 - for every finite subset of indices $i_1, i_2, \dots, i_k$
 - and every choice of real numbers $x_1, \dots, x_k$
 - $P(X_{i_1} \leq x_1, \dots, X_{i_k} \leq x_k) = \prod_{j=1}^k P(X_{i_j} \leq x_j)$
- Pairwise independence is the much weaker condition when $k=2$

$$\forall x_1, x_2, \dots, x_n, \quad f_{X_1, X_2, \dots, X_n}(x_1, x_2, \dots, x_n) = f_{X_1}(x_1) f_{X_2}(x_2) \cdots f_{X_n}(x_n)$$

$$\forall (x_i, x_j), 1 \leq i \leq n, 1 \leq j \leq n, i \neq j: \quad f_{X_i, X_j}(x_i, x_j) = f_{X_i}(x_i) f_{X_j}(x_j)$$

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Example

- Take the sample space $\{1, 2, 3, 4\}$, each singleton equally likely.
 - $P(\{1\}) = \frac{1}{4}$
- Let $A = \{1, 2\}$, i.e., either the event $\{1\}$ or the event $\{2\}$
 - $P(A) = \frac{1}{2}$
 - Similarly, $B = \{1, 3\}$, $C = \{1, 4\}$ and $P(A) = P(B) = P(C) = \frac{1}{2}$.
- Consider the intersection event $AB = \{1\}$. $P(AB) = \frac{1}{4}$
 - A and B are considered independent since $\frac{1}{2} * \frac{1}{2} = \frac{1}{4}$
- However, the intersection event ABC
 - Has $P(ABC) = P(\{1\}) = \frac{1}{4} \neq P(A)P(B)P(C) = \frac{1}{8}$, so A, B, C are not mutually independent.

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Events in the Example

- Define indicator random variables
 - $X_A = \mathbf{1}_A$, $X_B = \mathbf{1}_B$, $X_C = \mathbf{1}_C$, where, for example,
 - $X_A(\omega) = \begin{cases} 1, & \omega \in A, \\ 0, & \omega \notin A. \end{cases}$
- Every pair is independent. For example,
 - $P(X_A = 1, X_B = 1) = \frac{1}{4} = P(X_A = 1)P(X_B = 1)$.
- But the three variables are not mutually independent because
 - $P(X_A = 1, X_B = 1, X_C = 1) = \frac{1}{4} \neq \left(\frac{1}{2}\right)^3$.

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Lack of Transitivity Example

- Independence is not transitive
- Suppose a system contains two physically independent components.
 - Event A: Component 1 fails today.
 - Event B: Component 2 fails today.
 - Event C: Component 1 works today.

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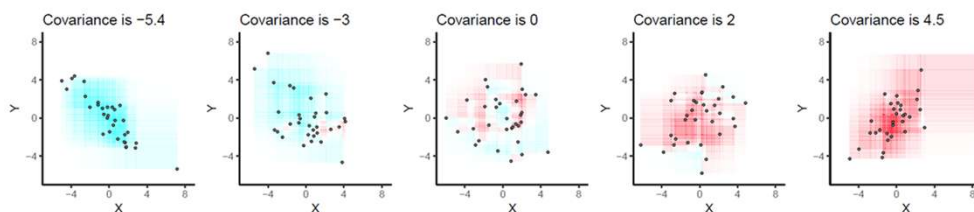
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Linearity and Covariance

$$\text{Cov}(X, Y) = E[(X - \mu_X)(Y - \mu_Y)]$$

- Expected values of sums: $E[X] + E[Y] = E[X+Y]$
- However, $\text{Var}[X] + \text{Var}[Y] \neq \text{Var}[X+Y]$
 - Consider $Y = X$
 - We need the concept of covariance
 - Think of pairs (x, y) to define red and blue
 - A point is red if the points "agree"

$$s_{xy} = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})$$



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Correlation Coefficient

- The strength of the relationship is measured by normalization
 - Unit free, bounded and easy to compare

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Correlation Coefficient

- The strength of the relationship is measured by normalization
 - Unit free, bounded and easy to compare
- Sample correlation coefficient r
 - $r > 0$ positively correlated
 - $r < 0$ negatively correlated
 - r close to 0 uncorrelated

$$r(X, Y) = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X) \text{Var}(Y)}}$$

$$r = \frac{\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sigma_x \sigma_y}$$

$$r_{xy} = \frac{s_{xy}}{s_x s_y}$$

$$r = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2} \sqrt{\sum_{i=1}^n (y_i - \bar{y})^2}}$$

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Correlation Is Not Causation

- Correlation does not necessarily imply causation
 - Consider: *High correlation between tobacco usage and lung cancer occurrence does not imply that smoking causes lung cancer*
- Sensitive to outliers

Let's go back to linearity

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Alternate Formula

$$E[(X - \mu_X)(Y - \mu_Y)]$$

$$\text{var}(X) = E[X^2] - (E[X])^2$$

- A formula similar to that of the mean: $\text{Cov}(X, Y) = E[XY] - E[X]E[Y]$

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Covariance and variance

$$\text{Cov}(X,Y) = E[XY] - E[X]E[Y]$$

$$\text{Cov}(X+Z, Y) = E[(X+Z)Y] - \uparrow$$

- Symmetric: $\text{Cov}(X,Y) = \text{Cov}(Y,X)$
- “Bilinear”: Linear in each argument separately
 - $\text{Cov}(aX,Y) = a \text{Cov}(X,Y)$
 - Recall: $\text{Var}(aX) = a^2 \text{Var}(X)$
 - $\text{Cov}(X+Z, Y) = \text{Cov}(X,Y) + \text{Cov}(Z,Y)$
- $\text{Cov}(X,X) = \text{Var}(X)$
- $\text{Var}(X+Y) = \text{Var}(X) + \text{Var}(Y) + 2\text{Cov}(X,Y)$

$$\text{Cov}(X+Y, X+Y) = \text{Cov}(X, X+Y) + \text{Cov}(Y, X+Y)$$

$$\text{Cov}(X,Y) = E[(X - \mu_X)(Y - \mu_Y)]$$